ECG Validation Report

Patient	184B27	Segment	1780171482351_VC2B008BF_184B27_ecg_seg1
Recording start	2026-05-30 20:04:42 UTC	Duration	115.44 s
Device	VitalPatch (single-lead)	Sampling rate	125.0 Hz -> 125.0 Hz
Pipeline	ecg_pipeline (agent_bridge.run_full_report)	Git commit	e0da1f7
Report generated	2026-08-02	Beats detected / analyzed	127 / 119

SIGNAL QUALITY

SQI score: 0.61 Rejection rate: 39.1% Usable signal: 60.9%

Baseline wander: YES Flatline: YES Clipping: NO

SQI windows: 23 total, 9 rejected (FLATLINE x2, BASELINE_WANDER x1, NO_QRS_IMPULSE_CHARACTER x6)

Heart rate: 118.1 bpm (median_nonflagged_rr, 106 of 126 intervals)

CRITICAL

Deciding rule

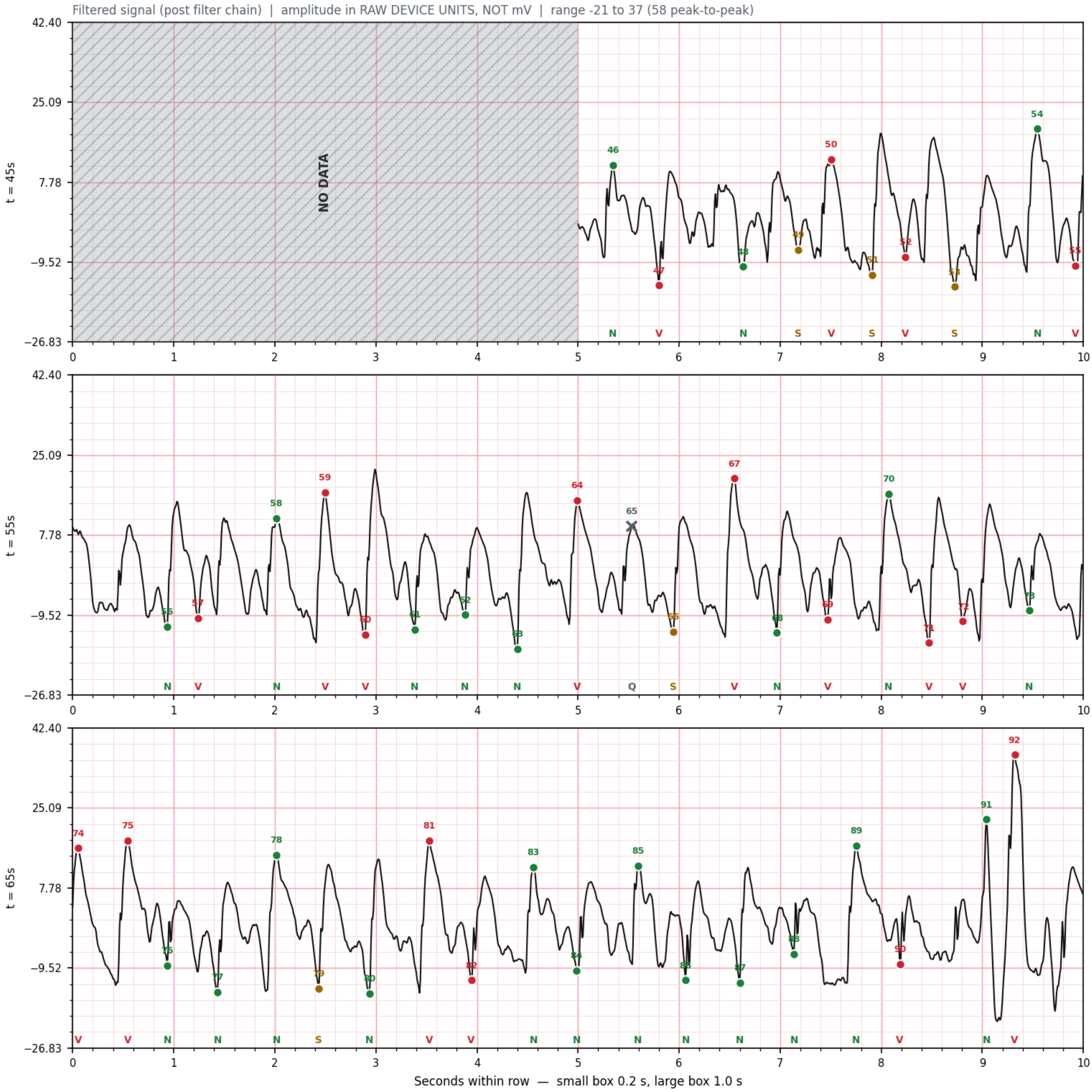
PVC burden > CRITICAL threshold — measured 27.559 vs threshold 20.0 -> FIRED

How to use this document

Page 2 shows the cleaned ECG with each detected beat marked and labelled. Page 3 lists what the classifier decided and how confident it was. Page 4 shows every rule the system evaluated. Page 5 is the form for your assessment — please complete it. Page 6 states what this system can and cannot do.

Cleaned ECG strip

Showing 30 s from t = 45.0s to 75.0s (the window with the most ectopic beats in this segment), in 3 rows of 10 s



Beat markers:

- N — Normal sinus beat
- V — Ventricular (PVC, ectopic)
- S — Supraventricular (PAC, junctional)
- F — Fusion beat
- X Q — Artifact / rejected by quality gate

Grey hatched regions = signal rejected by the quality gate (NO DATA — not recorded signal, and not asystole).
Small numbers above each marker are beat indices, matching the tables on pages 3 and 5.

Beat sequence in this window

N V N S V S V S N V N V N V V N N N V Q S V N V N V V N V V N N N S N V V N N N N N N N V N V

Full segment: 127 beats detected. Sequence above covers the displayed window only.

Class	Description	Count	% of detected	Avg conf.	Min conf.
N	Normal sinus beat	77	60.6%	0.897	0.398
V	Ventricular (PVC, ectopic)	35	27.6%	0.834	0.376
S	Supraventricular (PAC, junctional)	7	5.5%	0.554	0.388
F	Fusion beat	0	0.0%	—	—
Q	Artifact / rejected by quality gate	8	6.3%	N/A (rejected)	

S class F1 = 0.139 and F class F1 = 0.011 on the held-out validation set. Treat S and F counts as a screening signal only — not reliable.

Percentages are of ALL detected beats (including quality-rejected Q beats), matching the burden figures used by the rule engine.

Lowest-confidence beats (n=5)

#	Beat	Time (s)	Label	Confidence	Prev	Next	Runner-up
1	52	53.24	V	0.3760	S	S	N 0.374
2	79	67.43	S	0.3879	N	N	V 0.361
3	51	52.91	S	0.3973	V	V	N 0.307
4	68	61.97	N	0.3983	V	V	V 0.368
5	41	42.69	S	0.4941	N	V	V 0.420

Why the model chose these labels (SHAP attributions)

Exact TreeSHAP from the frozen model, in margin (log-odds) space. These explain the MODEL's decision — not that the decision is correct.

Beat 52 at 53.24s — predicted V (conf 0.376)

rr_pre_ms SHAP +1.027 (toward V) value 328.000 — RR interval before this beat (prematurity)
wavelet_d3_9 SHAP +0.569 (toward V) value 0.221 — wavelet shape coefficient
wavelet_d3_8 SHAP +0.519 (toward V) value -0.261 — wavelet shape coefficient

Beat 79 at 67.43s — predicted S (conf 0.388)

wavelet_d4_5 SHAP +0.616 (toward S) value 1.799 — wavelet shape coefficient
rr_pre_ms SHAP +0.456 (toward S) value 416.000 — RR interval before this beat (prematurity)
area_ratio_pre_post SHAP -0.308 (away from S) value 0.230 — pre-R vs post-R waveform area (QRS asymmetry)

Beat 51 at 52.91s — predicted S (conf 0.397)

local_hrv_ms SHAP -0.947 (away from S) value -80.000 — change in RR across this beat (short-long pattern)
rr_pre_ms SHAP +0.527 (toward S) value 408.000 — RR interval before this beat (prematurity)
wavelet_a4_9 SHAP +0.229 (toward S) value 5.212 — wavelet shape coefficient

Beat 68 at 61.97s — predicted N (conf 0.398)

rr_pre_ms SHAP -2.662 (away from N) value 424.000 — RR interval before this beat (prematurity)
local_hrv_ms SHAP +0.595 (toward N) value 80.000 — change in RR across this beat (short-long pattern)
amplitude_range SHAP +0.227 (toward N) value 3.844 — peak-to-peak amplitude of the beat window (QRS size)

Beat 41 at 42.69s — predicted S (conf 0.494)

rr_pre_ms SHAP +0.878 (toward S) value 416.000 — RR interval before this beat (prematurity)
wavelet_d3_7 SHAP -0.284 (away from S) value 0.430 — wavelet shape coefficient
wavelet_a4_9 SHAP +0.276 (toward S) value 3.193 — wavelet shape coefficient

Rule trace — every rule the system evaluated

Rule	Threshold	Provenance	Measured	Fired	Sets level
PVC burden > CRITICAL threshold	20.0	PROJECT-DEFINED	27.559	YES	CRITICAL
VT run count > 0 (a run = >=3 consecutive V b…	0	MIXED	0	no	CRITICAL
PVC burden > HIGH threshold	10.0	PROJECT-DEFINED	27.559	YES	HIGH
PAC burden > HIGH threshold OR AFib burden > …	15.0 / 30.0	MIXED	5.512 / 100.0	YES	HIGH
Sustained HRV suppression: SDNN < threshold	20.0	PROJECT-DEFINED	164.351	no	MEDIUM
NEWS2 safety override (>= critical threshold)	7	CLINICAL LITERATURE	4	no	CRITICAL
qSOFA safety override (>= high threshold)	2	CLINICAL LITERATURE (threshold only)	1	no	HIGH

The deciding rule is shown in red. Rules are evaluated in severity order, first match wins.

Threshold provenance notes

- Not externally validated.
- The >=3-beat definition of a run is standard clinical usage; escalating ANY run straight to CRITICAL is project-defined.
- Burden cutoffs (PAC 15%, AFib 30%) are project-defined. The underlying AFib detector threshold (RR-CV > 0.10) IS validated against LTAFDB (84 records, 449,749 annotated windows).
- In-code description is 'conservative low-HRV cutoff'. Not from clinical literature.
- Royal College of Physicians NEWS2 (2017).
- Seymour et al. JAMA 2016 sets the >=2 threshold. This system's score is a 1-of-3-criteria proxy, not a true qSOFA -- see page 6.

Vitals used in this assessment

Vital	Value	NEWS2 score	Source
Heart rate (bpm)	115.8	2	vitalpatch_col1
Respiratory rate (/min)	23.8	2	vitalpatch_col2
Temperature (C)	NOT AVAILABLE	—	no valid readings in vitals file
SpO2 (%)	NOT AVAILABLE	—	VitalPatch has no SpO2 sensor
Systolic BP (mmHg)	NOT AVAILABLE	—	VitalPatch has no BP sensor
Consciousness (AVPU)	NOT AVAILABLE	—	AVPU/consciousness has no input path on this device

NEWS2: 4 (2 of 6 components) qSOFA: 1 (1 of 3 components)

Unavailable components contribute 0 and are listed, never imputed — both scores are LOWER BOUNDS, not complete scores. Time gap to nearest vitals: 0.0 min (match: interval_containment).

Please review the ECG strip on page 2 and answer the questions below. Your answers will be used to validate the automated analysis. This form is fillable digitally in Acrobat Reader, Firefox, Edge or Preview, or may be completed by hand.

SECTION A — Overall assessment

Q1. Does the ECG strip show the arrhythmia described?

☐ Yes, clearly visible
☐ Partially — some findings present
☐ No — I do not see this arrhythmia
☐ Cannot assess — signal quality too poor

Q2. Is the CRITICAL verdict clinically justified?

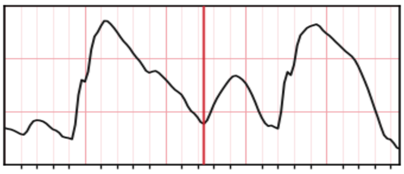
☐ Yes — I would act on this alert
☐ Probably — warrants monitoring
☐ No — this appears to be a false alarm
☐ Uncertain — need more information

Q3. Is the 20% PVC burden threshold appropriate for triggering a CRITICAL alert?

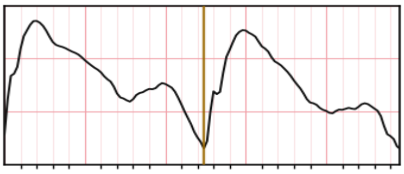
☐ Yes — appropriate threshold
☐ Too sensitive — should be higher, suggest: ____ %
☐ Not sensitive enough — should be lower, suggest: ____ %

SECTION B — Beat-by-beat review

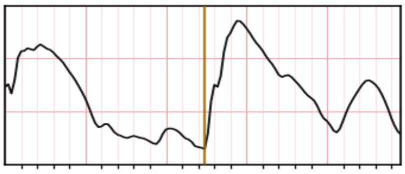
These are the beats the model was least sure about — where your judgement is worth the most. Each strip is 1 second centred on the beat.



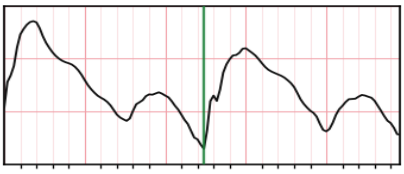
Beat 52 t=53.24s
Pipeline: V (conf 0.376)
☐ N ☐ S ☐ V ☐ Q ☐ ?
agree / correct label / unsure(?)



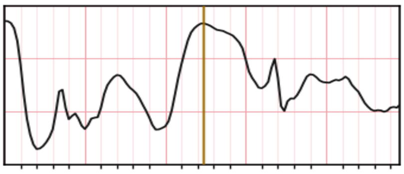
Beat 79 t=67.43s
Pipeline: S (conf 0.388)
☐ N ☐ S ☐ V ☐ Q ☐ ?
agree / correct label / unsure(?)



Beat 51 t=52.91s
Pipeline: S (conf 0.397)
☐ N ☐ S ☐ V ☐ Q ☐ ?
agree / correct label / unsure(?)



Beat 68 t=61.97s
Pipeline: N (conf 0.398)
☐ N ☐ S ☐ V ☐ Q ☐ ?
agree / correct label / unsure(?)



Beat 41 t=42.69s
Pipeline: S (conf 0.494)
☐ N ☐ S ☐ V ☐ Q ☐ ?
agree / correct label / unsure(?)

SECTION C — Free text

What is missing from this report that you would need to make a clinical decision?

What would need to change before you would use this system clinically?

Reviewer **Specialty** **Date**

Signature ______________________________

What this system CAN do

- Detect R-peaks. Validated against the device's own firmware-reported RR intervals: 93.3% of segments agree within 20 ms; mean heart-rate difference 0.68 bpm; n = 2,651 segments against 4,364,472 reference intervals.
- Flag elevated PVC burden above a configurable threshold.
- Identify rhythm patterns — bigeminy, trigeminy, VT runs, RR-irregularity suggestive of AF.
- Provide a complete, explainable rule trace for every verdict it produces.
- Compute a partial early-warning score from the vitals the device does measure.

What this system CANNOT do

- Diagnose arrhythmias. This is decision support only.
- Reliably classify supraventricular (S / PAC) beats — held-out F1 = 0.139.
- Classify fusion (F) beats — held-out F1 = 0.011, essentially unsolved.
- Compute a complete NEWS2. SpO2 and blood pressure are not measured by this device, so the score is a lower bound computed from a subset of components.
- Compute a true qSOFA. Two of its three criteria (systolic BP, altered mentation) have no input on this device; the reported figure is a 1-of-3 proxy that can never reach the published threshold of 2.
- Establish accuracy on this device. The classifier was trained and validated on MIT-BIH (2-lead, different population). No clinician-annotated VitalPatch labels exist — which is precisely what this review is for.
- Replace clinical judgement.

Known limitations of THIS segment

- Beat classifier is the frozen production XGBoost model (five_class_xgb.json). DS2 held-out per-class F1: V=0.826 (reliable), S=0.139 (LOW-CONFIDENCE), F=0.011 (LOW-CONFIDENCE, essentially unsolved). Any finding driven primarily by S or F counts is a screening flag, not a diagnosis.
- NEWS2 is scored from only the components VitalPatch measures (HR, respiratory rate, skin temperature). SpO2, blood pressure and consciousness are not measurable on this hardware, contribute 0, and are listed in missing_components -- the total is a LOWER BOUND, not a complete NEWS2. qSOFA is a 1-of-3-criteria proxy whose maximum value is 1, below its own threshold of 2, so it can never fire on device data alone. When no vitals were matched, safety_overrides are reported as not-evaluated rather than fabricated.
- 39% of this recording was discarded by the quality gate, leaving 61% usable. Detected beat counts and any rate derived from them cover only the surviving signal.
- 23 of 119 classified beats were below 0.70 confidence.